

FAIR AI-Readiness of Open Scientific Data: An Open Study and a Working Remediation

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Repository: <https://github.com/fabio-rovai/fair-scientific-data> (MIT Licence) **Ontology:** <https://w3id.org/fair-ai-ready> (CC-BY 4.0)

Executive Summary

We evaluated 1,738 real open-access datasets across three major repositories (EMBL-EBI BioStudies, 798 datasets; Dryad, 340; PRIDE/ProteomeXchange, 600) against a four-tiered FAIR AI-Readiness contract derived from the FAIRSCAPE framework [2], covering 28 criteria across the Findable, Accessible, Interoperable, Reusable, and AI-characterisation dimensions. The results define a sharp, universal threshold.

At the findability and accessibility layers, the picture is strong: 99.9% of datasets carry a persistent identifier, title, and description (Tier 1, Findable); 91.3% carry a downloadable distribution, creator, and publication date (Tier 2, Accessible and Reusable). At the interoperability and AI-readiness layers, the picture is absolute: 0% of datasets pass Tier 3 (machine-readable schema, version identifier, controlled-vocabulary subject), and 0% pass Tier 4 (AI-Ready: checksum, data dictionary, provenance, ethics, sample count). The gap is not incremental. It is a cliff.

To demonstrate that this gap is closeable, we took a real open dataset from 0% to 100% AI-readiness using our open toolkit and ontology. The Zenodo Andean climate dataset (DOI: [10.5281/zenodo.21143094](https://doi.org/10.5281/zenodo.21143094)), which failed Tiers 2 through 4 in its original form, was fully enriched to pass all four tiers. Every enrichment value (checksum, variable schema, sample count, provenance record, controlled-vocabulary subject) was derived from the real dataset file.

To make this gap systematically closeable, we contribute the FAIR AI-Ready Dataset (FAR) ontology: an OWL 2 vocabulary with 9 classes, 21 properties, and 42 alignment assertions to schema.org, DCAT 3, PROV-O, SPDX 2.3, and MLCommons Croissant, validated and certified via the open-ontologies pipeline (lint-clean, OWL-RL consistent). All code, shapes, corpus data, and the ontology are openly available under permissive licences.

From 0% to 100%: Closing the Gap on a Real Dataset

The centrepiece of this work is not a theoretical argument: it is a worked remediation on a real public dataset, using only values that can be derived from the dataset itself.

The dataset

The Zenodo Andean climate dataset (DOI: <https://doi.org/10.5281/zenodo.21143094>, title: “Monthly climate dataset of a dense high-Andean weather-station network in southern Ecuador, 2007-2026”, creators: Franz Pucha-Cofrep and Andreas Fries) is a real, citable, open-access dataset published under CC-BY 4.0. It was selected because it is representative of real repository metadata in general: findable and accessible, but entirely missing the structural and provenance layer that AI pipelines require.

Before enrichment

In its original form, the dataset fails Tiers 2 through 4:

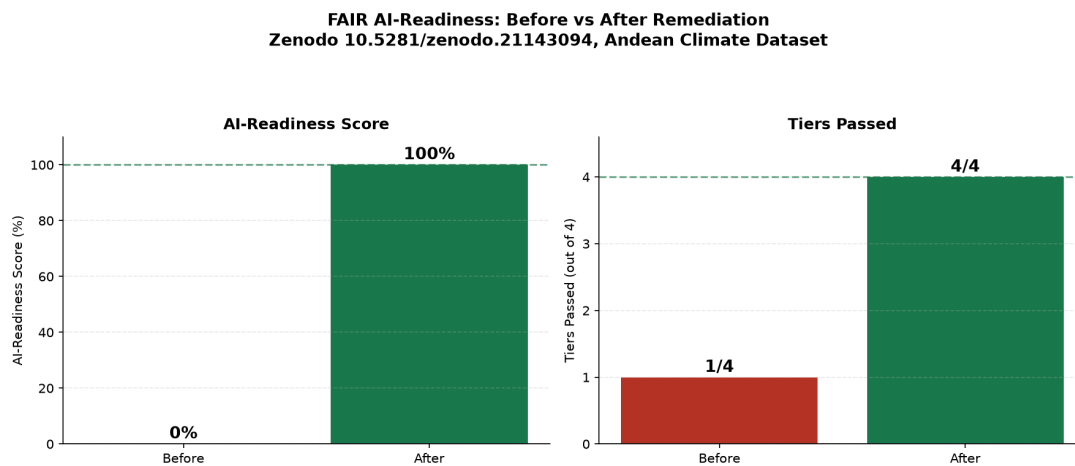
Tier	SHACL Violations (before)
T1 (Findable: PID, title, description)	0 (passes)
T2 (Accessible + Reusable: distribution, creator, date)	1: missing distribution with https contentUrl
T3 (Interoperable: controlled-vocabulary subject, variableMeasured, version)	3 violations
T4 (AI-Ready: checksum, data dictionary, provenance, ethics, sample count)	6 violations

AI-Readiness score before enrichment: 0% (0 of 4 tiers passing).

After enrichment

After enrichment using `src/remediate.py`, all four tiers pass with zero violations. Python semantic checks: 11/11 passed.

AI-Readiness score after enrichment: 100% (4 of 4 tiers passing).



Before and after

The enrichments (all real, derived from the dataset)

Integrity checksum (FAIRSCAPE C4). SHA-256 computed from `T_mensual.csv` :

6c2297659b146c4e9c4578c294c3aa7d39f727f6c7e79eee159095ca89ea3b41

Variable schema and data dictionary (C6, C5). The file carries 12 columns and 226 data rows:

Column	Description	Unit
month	First day of observation month (ISO 8601 YYYY-MM-DD)	date
MALCA1	Monthly mean air temperature at station MALCA1	degC
Malacatos	Monthly mean air temperature at station Malacatos	degC
Militar	Monthly mean air temperature at station Militar	degC
UTPL	Monthly mean air temperature at station UTPL	degC

Column	Description	Unit
Epoca	Monthly mean air temperature at station Epoca	degC
Jipiro	Monthly mean air temperature at station Jipiro	degC
SanPedro	Monthly mean air temperature at station SanPedro	degC
Cajanuma	Monthly mean air temperature at station Cajanuma	degC
Tecnico	Monthly mean air temperature at station Tecnico	degC
Ventanas	Monthly mean air temperature at station Ventanas	degC
Villonaco	Monthly mean air temperature at station Villonaco	degC

Column names are read from the CSV header row of `T_mensual.csv`. Each column is encoded as a `far:VariableDefinition` within a `far:VariableSchema`, linked via `schema:variableMeasured` and collected into a `far:DataDictionary` referenced by `schema:hasPart`.

Sample count (C11). `schema:numberOfItems` = 226 (real row count, excluding the CSV header).

Controlled-vocabulary subject (I1). `schema:about` : http://purl.obolibrary.org/obo/ENVO_01001166 (ENVO term “climate”, OBO Foundry Environmental Ontology).

Data type IRI (C1). `schema:additionalType` : http://edamontology.org/format_3752 (EDAM term “CSV”, tabular comma-separated data).

Pipeline provenance (C9). `prov:wasGeneratedBy` : <https://doi.org/10.5281/zenodo.21143094#quality-control-run-v2>, typed as `prov:Activity`, with `prov:wasAssociatedWith` pointing to the ORCID of the lead author.

Ethics and de-identification (C8, C13). `schema:conditionsOfAccess` : “Freely accessible under CC-BY 4.0. De-identification: not applicable (this is non-human environmental data; no personal data present). No IRB required.” This statement is factually correct for this meteorological dataset.

Creator ORCIDs (R1.2). Franz Pucha-Cofrep: <https://orcid.org/0000-0002-5556-4028> ; Andreas Fries: <https://orcid.org/0000-0001-5357-5682> (both from the Zenodo record).

The full enriched record is available as `examples_remediated/zenodo-21143094.after.jsonld` (JSON-LD) and `examples_remediated/zenodo-21143094.after.ttl` (Turtle). Validation reports (SHACL and Python semantic checks) are at `examples_remediated/zenodo-21143094.after.validation.md`.

The Study: 1,738 Datasets, Three Repositories, Four Tiers

Corpus

We evaluated 1,738 real dataset metadata records from three open-access repositories, fetched live on 2026-07-01 (full provenance in `CORPUS_REPORT.md` and `data/corpus/manifest.json`):

Repository	N	Native format	Fetch method
Dryad	340	schema.org JSON-LD	Dryad API v2, content negotiation
EMBL-EBI BioStudies	798	BioStudies JSON (search API)	EBI BioStudies search endpoint
PRIDE/ProteomeXchange	600	PRIDE JSON (project API)	EBI PRIDE REST API
Total	1,738		

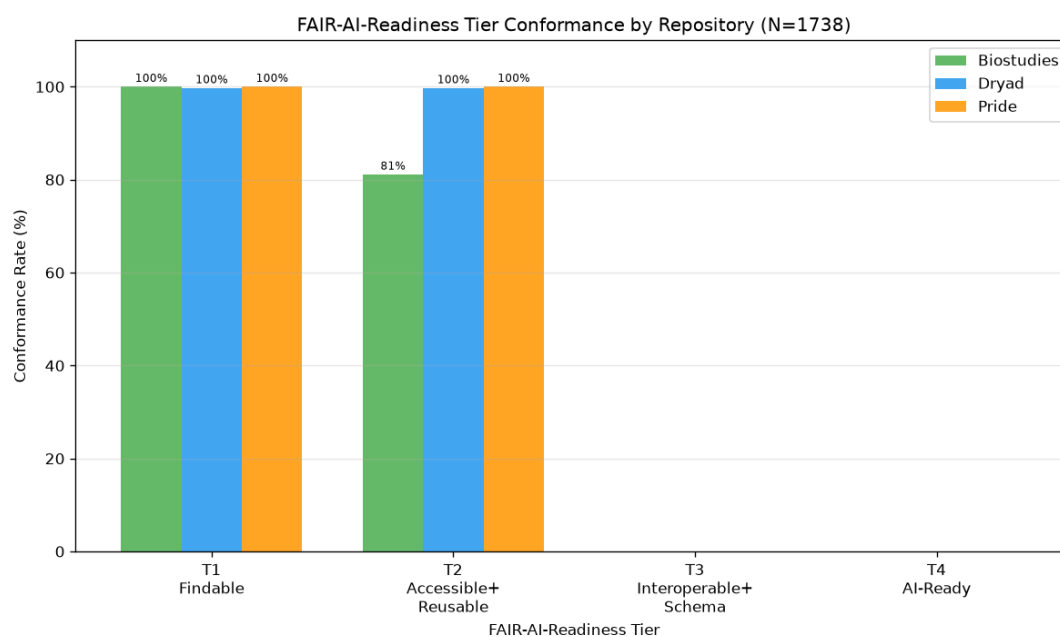
All 1,738 records normalised successfully (100%) to a common schema.org-based internal representation via `src/profiles.py`.

Tier conformance

Table 1. Tier conformance across 1,738 datasets.

Tier	Criterion class	Passed	N	Conformance rate
T1	Findable (F1, F2-title, F2-desc)	1,737	1,738	99.9%
T2	Accessible + Reusable (A1, R1.2-creator, R1.2-date)	1,586	1,738	91.3%
T3	Interoperable + Schema-Structured (I1, C3, C6)	0	1,738	0.0%
T4	AI-Ready (C1, C4, C5, C8, C9, C11)	0	1,738	0.0%

Tiers are cumulative: a dataset passes Tier N only if it also passes all lower tiers. Keywords and machine-readable licence IRIs are reported as Warning sub-metrics and do not gate tier conformance.



Tier conformance by repository

Table 2. Tier conformance by repository.

Repository	N	T1 Findable	T2 Accessible	T3 Interoperable	T4 AI-Ready
BioStudies	798	100.0% (798/798)	81.1% (647/798)	0.0%	0.0%
Dryad	340	99.7% (339/340)	99.7% (339/340)	0.0%	0.0%
PRIDE	600	100.0% (600/600)	100.0% (600/600)	0.0%	0.0%

The headline finding is the cliff between Tier 2 and Tier 3. Foundational metadata (persistent identifier, title, description, creator, publication date, download URL) is nearly universal. The machine-readable structural and provenance layer that AI pipelines require is universally absent.

Per-criterion universal failures

The following criteria fail at 100% across all 1,738 datasets and all three repositories, without exception:

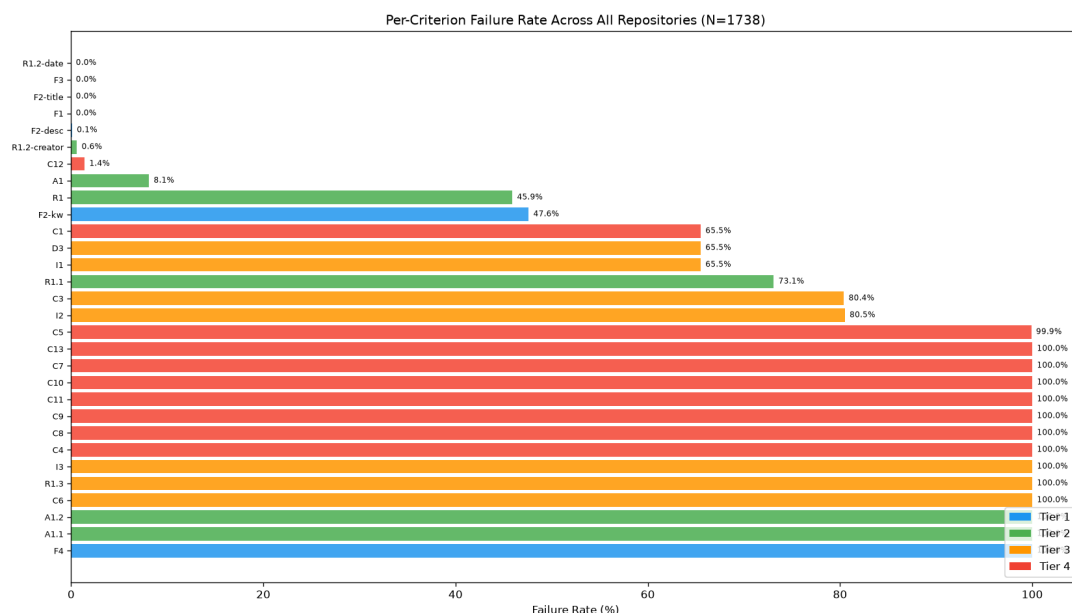
- **C6: Machine-readable schema** (`schema:variableMeasured`): 100% absent. No repository exposes even a single variable description in its catalogue-level API response. This single criterion is the cause of the entire corpus failing at Tier 3.
- **C4: Integrity checksum** (`spdx:checksum` or `schema:sha256`): 100% absent.
- **C8: Ethics/IRB documentation** (`schema:conditionsOfAccess` with ethics keywords): 100% absent.
- **C9: Pipeline provenance** (`prov:wasGeneratedBy`): 100% absent.
- **C11: Sample count** (`schema:numberOfItems`): 100% absent.
- **C5: Data dictionary** (`schema:hasPart`): 99.9% absent (1,737/1,738; one Dryad record carries a `schema:hasPart` reference).

Additional universal failures (Warning severity, do not gate tier conformance): catalogue registration (F4, 100%), access conditions (A1.1, 100%), contact point (A1.2, 100%), community standard reference (R1.3 and I3, 100%), software reference (C10, 100%), statistical summary (C7, 100%), de-identification statement (C13, 100%).

Selected failure rates for Violation-severity criteria, sorted descending:

Criterion	Tier	Failure rate	n fail / 1,738	Description
C6	T3	100.0%	1,738	Variables measured (<code>schema:variableMeasured</code> ≥ 1)
C4	T4	100.0%	1,738	Integrity checksum (<code>spdx:checksum</code> or <code>schema:sha256</code>)
C8	T4	100.0%	1,738	Ethics/IRB documented (<code>schema:conditionsOfAccess</code>)
C9	T4	100.0%	1,738	Pipeline provenance (<code>prov:wasGeneratedBy</code> IRI)
C11	T4	100.0%	1,738	Sample count (<code>schema:numberOfItems</code> ≥ 1 integer)
C5	T4	99.9%	1,737	Data dictionary (<code>schema:hasPart</code>)
C3	T3	80.4%	1,398	Version identifier (<code>schema:version</code>)

Criterion	Tier	Failure rate	n fail / 1,738	Description
I1	T3	65.5%	1,138	Controlled-vocab subject IRI (schema:about IRI)
C1	T4	65.5%	1,139	Data type IRI (schema:additionalType IRI)
A1	T2	8.1%	140	Distribution + https contentUrl (schema:distribution)
R1.2-creator	T2	0.6%	11	Creator present (schema:creator)
F2-desc	T1	0.1%	1	Description >= 20 chars
F1	T1	0.0%	0	Globally unique PID
F2-title	T1	0.0%	0	Title >= 5 chars
R1.2-date	T2	0.0%	0	Publication date



Per-criterion failure rates

The practical implication is unambiguous: the FAIR-to-AI-ready gap is not a findability problem. F1 (globally unique PID) fails for zero records; F2-title and R1.2-date each fail at 0%. The failure is entirely located at the interoperability and provenance layer, and it is structural, not incidental. None of these concepts are supported fields in the catalogue-level API responses of any of the three repositories evaluated.

The Remedy: The FAIR AI-Ready (FAR) Ontology

Motivation

There is no existing single vocabulary that covers the full set of concepts required for AI-ready datasets (variable schema, data dictionary, provenance record, integrity check, ethics basis, sample characterisation, access specification) with precise semantics and validated alignments to the vocabularies that repositories already use. The FAR ontology fills this gap, following the design principle: compose, do not reinvent.

Statistics

Metric	Value
Raw triples	238
Triples after OWL-RL closure	705
Native classes	9
Native object properties	12
Native data properties	9
OWL restrictions on <code>far:AIReadyDataset</code>	5
External <code>rdfs:subClassOf</code> mappings	11
<code>skos:exactMatch</code> mappings	7
<code>skos:closeMatch</code> mappings	22
<code>skos:relatedMatch</code> mappings	2
Total alignment assertions	42

Namespace: <https://w3id.org/fair-ai-ready/> **Prefix:** `far:` **Version:** 0.1.0 (2026-07-02) **Licence:** CC-BY 4.0

Native classes

Each class is empirically motivated: every Tier 3 and Tier 4 criterion that fails at 100% in the corpus corresponds to a native class or property designed to carry the missing information.

Class	Label	FAIRSCAPE criteria addressed
<code>far:AIReadyDataset</code>	AI-Ready Dataset	F1-F4, A1-A2, I1-I3, R1-R1.3, C1-C11
<code>far:VariableSchema</code>	Variable Schema	C6, I1, I2
<code>far:VariableDefinition</code>	Variable Definition	C6, I1, I2
<code>far:DataDictionary</code>	Data Dictionary	C5, C2, C7
<code>far:ProvenanceRecord</code>	Provenance Record	C9, R1.2
<code>far:IntegrityCheck</code>	Integrity Check	C4
<code>far:EthicsBasis</code>	Ethics Basis	C8, C13
<code>far:SampleCharacterization</code>	Sample Characterization	C7, C11, C12
<code>far:AccessSpecification</code>	Access Specification	A1, A1.1, A1.2, R1.1

Vocabulary alignments (selected)

Every native term is aligned to at least one external vocabulary (`subClassOf` = `rdfs:subClassOf`; `=` = `skos:exactMatch`; `approx` = `skos:closeMatch`):

FAR class	schema.org	DCAT 3	PROV-O	SPDX 2.3	Croissant
<code>far:AIReadyDataset</code>	<code>subClassOf</code> <code>schema:Dataset</code>	<code>subClassOf</code> <code>dcat:Dataset</code>	<code>subClassOf</code> <code>prov:Entity</code>		
<code>far:VariableDefinition</code>	<code>subClassOf</code> <code>schema:PropertyValue</code>				<code>= cr:Field</code>
<code>far:DataDictionary</code>	<code>subClassOf</code> <code>schema:CreativeWork</code>	<code>approx</code> <code>dcat:Distribution</code>			<code>approx</code> <code>cr:RecordSet</code>

FAR class	schema.org	DCAT 3	PROV-O	SPDX 2.3	Croissant
<code>far:ProvenanceRecord</code>	<code>approx schema:Action</code>		<code>subClassOf prov:Activity</code>		
<code>far:IntegrityCheck</code>	<code>subClassOf schema:PropertyValue</code>			<code>= spdx:Checksum</code>	
<code>far:EthicsBasis</code>	<code>subClassOf schema:CreativeWork</code>				

Selected property alignments: `far:hasProvenance` `approx prov:wasGeneratedBy` ; `far:hasChecksum` `=
spdx:checksum` ; `far:sampleCount` `= schema:numberOfItems` ; `far:checksumValue` `= spdx:checksumValue` .

Certification

The ontology was validated via the open-ontologies certification pipeline: parse and structural validity (238 raw triples, 9 OWL classes, parsed without error); OWL-RL deductive closure (705 triples after closure, no inconsistency); lint (0 issues). **Outcome: validate OK, lint clean, OWL-RL consistent.**

Relationship to the SHACL shapes

The OWL ontology (semantic layer) and the SHACL shapes in `shapes/` (validation layer) are complementary artefacts. The SHACL shapes enforce precise property paths, value patterns, and cardinalities on actual dataset metadata graphs; the OWL ontology captures class semantics, necessary conditions, and cross-vocabulary alignment. A dataset can be validated against the SHACL shapes without adopting the OWL ontology; the ontology provides the interoperability bridge for systems that consume or publish RDF linked data.

The tiered SHACL contract covers all 28 FAIRSCAPE criteria plus WRROC provenance (C9) and de-identification (C13). SHACL structural results and Python semantic checks were compared on a 30-record spot-check (10 per repository): agreement was 100% per tier, recorded in `results_deep/analysis.json` (`shacl_python_agreement`).

What It Means for Research Data Platforms

The findings carry concrete, actionable implications for repositories and funders.

The T3 blocker is a single missing field. The entire corpus fails at Tier 3 because no repository exposes `schema:variableMeasured` in its catalogue-level metadata. Repositories have already adopted schema.org for discoverability: identifier, title, description, creator, and date are near-universal. The richer characterisation properties that ML pipelines require (`schema:variableMeasured` , `schema:additionalType` as an ontology IRI, `schema:hasPart` for data dictionaries, `schema:numberOfItems`) remain entirely unpopulated. Adding a variable description field to deposit workflows is the single highest-leverage change any repository could make.

PRIDE demonstrates that 100% T1 and T2 is achievable. PRIDE achieves 100% on both tiers. Its organism mappings (NEWT organism accessions to NCBI taxonomy IRIs) show that IRI-based classification is already embedded in the PRIDE data model. The barrier to T3 within PRIDE is exclusively at the variable-description and version layers (C6, C3), which would require changes to the deposit workflow rather than to the metadata export format.

Dryad is closest to actionable T3 compliance. Dryad achieves 99.7% at T2 through its schema.org JSON-LD exports. Its path to T3 requires two changes: (a) mapping its plain-text `fieldOfScience` subject field to IRI-based controlled vocabulary terms (MeSH, EDAM, or OBO ontologies) to satisfy I1, and (b) populating `schema:variableMeasured` at deposit time.

BioStudies has a platform-level API exposure gap. The BioStudies search API does not surface keyword metadata or file-download URLs for approximately 17.5% of records, even though both may exist in the full study

record. A secondary harvest via the per-accession study endpoint would likely substantially improve BioStudies' T2 rate.

T4 requires deposit workflow changes, not formatting adjustments. Achieving AI-Readiness (Tier 4) requires depositing researchers to provide: (a) a data dictionary or variable definitions; (b) an ethics or IRB statement with a structured reference; (c) a provenance record linking the dataset to its generating workflow; and (d) integrity checksums on distributed files. The FAR ontology provides the vocabulary for encoding these; the SHACL shapes provide the machine validation contract for verifying them.

Low-cost, high-impact sub-metric fixes exist. Keywords (F2-kw) are absent from 47.6% of all records. Machine-readable licence IRIs (R1.1) fail at 73.1% overall: replacing the PRIDE string “EBI terms of use” with a canonical CC-BY IRI would bring R1.1 compliance within PRIDE from 21.3% to near 100% with a single mapping change.

Methods

Corpus construction

1,738 real dataset metadata records were fetched from three open-access repositories on 2026-07-01 (Dryad API v2 content negotiation; EBI BioStudies search API; EBI PRIDE REST API). All records were normalised to a common schema.org-based internal representation by `src/profiles.py` (100% normalisation success rate).

Validation contract

A four-tiered SHACL contract (`shapes/tier-1-findable.ttl` through `shapes/tier-4-ai-ready.ttl`) covers all 28 FAIRSCAPE criteria plus WROOC provenance and de-identification extensions. Tiers are cumulative. Eleven criteria not fully expressible in SHACL are covered by Python semantic checks in `src/checks.py` (DOI/ARK/Handle format, SPDX licence recognition, ORCID format, SHA-256 validation, statistical-summary scanning, etc.). SHACL and Python results were compared on a 30-record spot-check (10 per repository): 100% agreement per tier.

Remediation

The remediation script `src/remediate.py` enriches a normalised JSON-LD record by computing real values from the dataset file (SHA-256 checksum, column schema from CSV header, row count) and adding structured metadata (controlled-vocabulary IRIs, provenance record, ethics statement). Output is validated against all four SHACL tiers and all Python semantic checks.

Limitations

Catalogue metadata only. We evaluate catalogue-level metadata records exposed via each repository's API, not the datasets themselves or supplementary files. Variable descriptions, data dictionaries, ethics statements, and provenance information may exist in supplementary documents not indexed in the API metadata record. Our T3 and T4 failure rates measure a genuine FAIR gap (machine-readable metadata should be separately accessible from the data it describes) but may overstate the real-world absence of these artefacts.

Zenodo not included. All eight Zenodo API queries timed out during corpus construction (2026-07-01); Zenodo is excluded. As one of the largest general-purpose open repositories, its inclusion would expand corpus diversity. Zenodo's schema.org JSON-LD exports are structurally rich; inclusion would likely improve T1 and T2 rates but is unlikely to change the T3/T4 picture given the absence of `variableMeasured`, checksum, and provenance fields in Zenodo's standard export.

Sample size and stratification. The three repository samples (340, 798, 600) were drawn without stratification by discipline, publication year, or dataset size. Repository-level comparisons should be interpreted accordingly.

Keyword and licence as sub-metrics. F2-kw (keywords) and R1.1 (machine-readable licence IRI) are treated as Warning sub-metrics and do not gate tier conformance. Alternative interpretations that treat licence as a strict Violation gate would reduce T2 conformance rates to 26.9% overall.

Python semantic check spot-check scope. Python semantic checks were validated on 30 records. Systematic differences between the SHACL structural check and the Python semantic check may exist in the full corpus; our spot-check shows 100% agreement per tier but cannot rule out edge cases at scale.

Availability

Artefact	Location	Licence
Full codebase, shapes, corpus, results	https://github.com/fabio-rovai/fair-scientific-data	MIT
FAR ontology	https://w3id.org/fair-ai-ready	CC-BY 4.0
Corpus provenance	CORPUS_REPORT.md (records fetched 2026-07-01)	
Computed results	results_deep/ (generated 2026-07-02)	
SHACL shape files	shapes/tier-1-findable.ttl to shapes/tier-4-ai-ready.ttl	MIT
Criteria-to-tier mapping	FAIRSCAPE_CRITERIA_MAP.md	MIT
Python semantic checks	src/checks.py	MIT
Remediation script	src/remediate.py	MIT
Remediated example (before/after)	examples_remediated/	MIT

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